How to Run the Applications

TRACE-AI-FR Interactive Local Setup Guide for Windows

Item	Description
Project	TRACE-AI-FR — Transparent Reporting of AI-related Claims in Evidence: A Forensic Reasoning Framework
Repository	https://github.com/prakasamv6/TRACE-AI-FR
Audience	Professor, reviewer, examiner
Runtime	Windows 10/11, 64-bit
Recommended Method	Use dist\TRACE-AI-FR.exe. No Python setup is required.

1. Purpose

This guide is written for reviewers who want to run TRACE-AI-FR without a programming setup. It explains how to open the application, what each input field means, how to test with the sample evidence image, and how to confirm that reports were created successfully.

TRACE-AI-FR examines forensic artifacts and produces structured reports about possible AI-platform usage indicators. The tool supports forensic reasoning; it does not automatically prove user intent or reconstruct the full session unless the evidence supports that conclusion.

2. Start Here: Reviewer Checklist

- ☐ I am using Windows 10 or Windows 11.
- ☐ I downloaded or cloned the TRACE-AI-FR project folder.
- ☐ I extracted the project ZIP if I downloaded it from GitHub.
- ☐ I can see the dist folder.
- ☐ I can see TRACE-AI-FR.exe inside dist.
- ☐ I have an evidence source ready.
- ☐ For the sample test, I extracted 2020JimmyWilson_E01_Image.zip first.

3. Important File Notes

File or Folder	Meaning	Recommended Use
dist\TRACE-AI-FR.exe or Dowloaded form Google Drive "TRACE-AI-FR.exe "	Packaged Windows application	Best option for professor/reviewer testing
2020JimmyWilson.E01	Sample image file package	Use the E01 file as evidence
demo_evidence	Small demo folder, if included	Optional quick folder test
QUICK_START.md	Short technical guide	Optional reference
ARCHITECTURE.md	Architecture details	Technical reference
BUILD_SUCCESS.md	Build notes	Technical reference

Sample image reminder: 2020JimmyWilson_E01 image file to test the application. Then select the extracted .E01 file and choose Mode = e01.

4. Choose Your Run Option

Goal	Best Option	Difficulty
Open the application normally	Double-click dist\TRACE-AI-FR.exe or Dowloaded form Google Drive "TRACE-AI-FR.exe "	Easy
Test the sample forensic image	2020JimmyWilson_E01 image file path , set Mode = e01	Easy
Test a folder export	Paste the folder path into Evidence, set Mode = mounted	Easy/Moderate

Goal	Best Option	Difficulty
Use the command line	Run the PowerShell commands in this guide	Optional
Run from the source code	Install Python dependencies	Technical only

5. Fastest Method: Open the App by Double-Clicking

1. Open File Explorer.
2. Open the TRACE-AI-FR project folder.
3. Open the dist folder.
4. Double-click TRACE-AI-FR.exe.
5. Confirm that the application window opens.

If Windows blocks the file, right-click TRACE-AI-FR.exe, choose Properties, select Unblock if shown, click OK, and open the file again.

6. Recommended Sample Test

Step 1 — Extract the sample image

- ☐ Locate 2020JimmyWilson_E01_Image.zip.
- ☐ Right-click the ZIP file.
- ☐ Choose Extract All.
- ☐ Extract it into a simple folder, for example:

```
C:\Evidence\2020JimmyWilson
```

Step 2 — Select the extracted E01 file

After extraction, use the extracted .E01 file as the Evidence path. Example:

```
C:\Evidence\2020JimmyWilson\2020JimmyWilson.E01
```

Step 3 — Use these app values

Field	What to Enter
Evidence	Path to the extracted 2020JimmyWilson.E01 file
Output	C:\TRACE-Output
Questions	Leave blank for the first test
Case Name	LOCAL-TEST-001
Examiner	Your name
Organization	Your school, agency, or organization
Enable carving	Unchecked for the first test
Mode	e01

Step 4 — Run and verify

- ☐ Confirm the Evidence path points to the extracted .E01 file.
- ☐ Confirm Mode is e01.
- ☐ Click Run Analysis.
- ☐ Wait for completion.
- ☐ Open the output folder.
- ☐ Open the generated report.

7. Application Input Field Guide

Field	Required?	Plain-English Meaning	First-Test Recommendation
Evidence	Yes	File, ZIP, or folder the app will analyze	Use the extracted sample .E01 file
Output	No	Folder where reports will be saved	Use C:\TRACE-Output
Questions	No	Optional .docx file with case questions	Leave blank
Case Name	No	Name for the case/test run	Use LOCAL-TEST-001
Examiner	No	Name of the person running the test	Enter your name
Organization	No	School, company, or agency name	Enter your organization
Enable carving	No	Turns artifact carving on/off	Leave unchecked first
Mode	No	Tells the app how to treat the input	Use e01 for the sample image

Refer to the following Fig. 1 for reference.

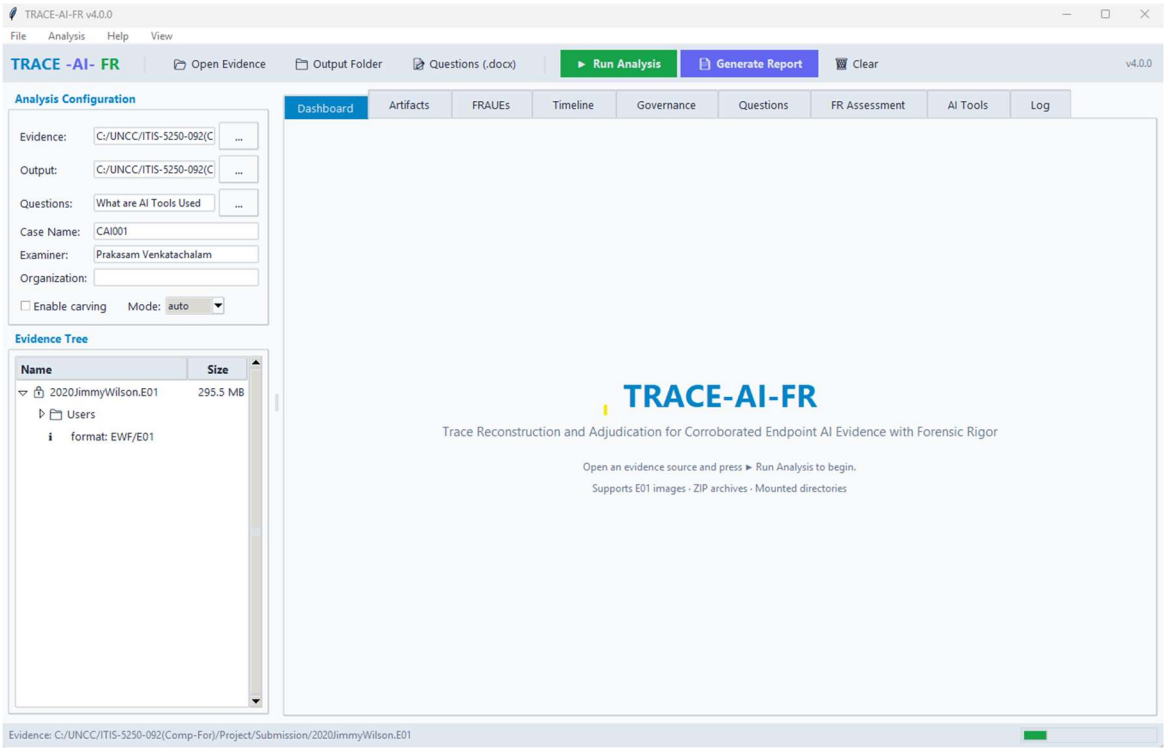


Fig - 1

Mode selection helper

Mode	Choose This When
auto	You are unsure
e01	You are analyzing an E01 forensic image
mounted	You are analyzing a normal extracted folder
zip	You are analyzing a ZIP archive

8. Optional PowerShell Commands

Use these only if you prefer command-line execution.

Open the app

```
cd dist
.\TRACE-AI-FR.exe
```

Analyze the sample E01 image

```
cd dist
.\TRACE-AI-FR.exe analyze `
  --evidence "C:\Evidence\2020JimmyWilson\2020JimmyWilson.E01" `
  --output "C:\TRACE-Output" `
  --case-id "LOCAL-TEST-001"
```

Test demo folder, if present

```
.\dist\TRACE-AI-FR.exe analyze `
  --evidence demo_evidence `
  --output demo_output_local `
  --case-id LOCAL-TEST-001
```

Show tool information

```
cd dist
.\TRACE-AI-FR.exe info
```

9. Output Verification

Look in the output folder for files similar to these:

Output	Purpose
*_report.docx	Word report for review/submission
*_report.html	Browser-readable report
*_report.md	Markdown report
*_findings.json	Structured findings
*_governance_record.json	Governance/traceability record
exhibits folder	Supporting artifacts, if generated

Verification checklist

- ☐ I found the output folder.
- ☐ I found at least one report file.
- ☐ I opened the report successfully.
- ☐ The case name appears correctly.
- ☐ Examiner/organization details appear if entered.

10. Generate Report Dialog

After analysis, click Generate Report if available. The dialog may ask for:

Dialog Field	Example
Forensic Examination Name	TRACE-AI-FR Local Validation Test
Forensic Examination in the Matter of	Sample E01 Evidence Review
Examiner Name	Your name

At least one of the first two fields should be filled. If both are blank, the app may show a warning.

11. Troubleshooting

Problem	What It Usually Means	Fix
App does not open	Windows blocked the executable	Right-click .exe → Properties → Unblock
App still does not open	Permission issue	Right-click → Run as administrator
Evidence missing warning	Evidence path is blank or wrong	Paste the full path again
Sample E01 not found	ZIP was not extracted	Extract 2020JimmyWilson_E01_Image.zip first
Folder test fails	Wrong mode	Set Mode to mounted
ZIP test fails	Wrong mode	Set Mode to zip
Not sure which mode to use	Input type unclear	Use auto
No reports found	Output folder is wrong	Check the Output field
No Questions file	Questions file is optional	Leave Questions blank

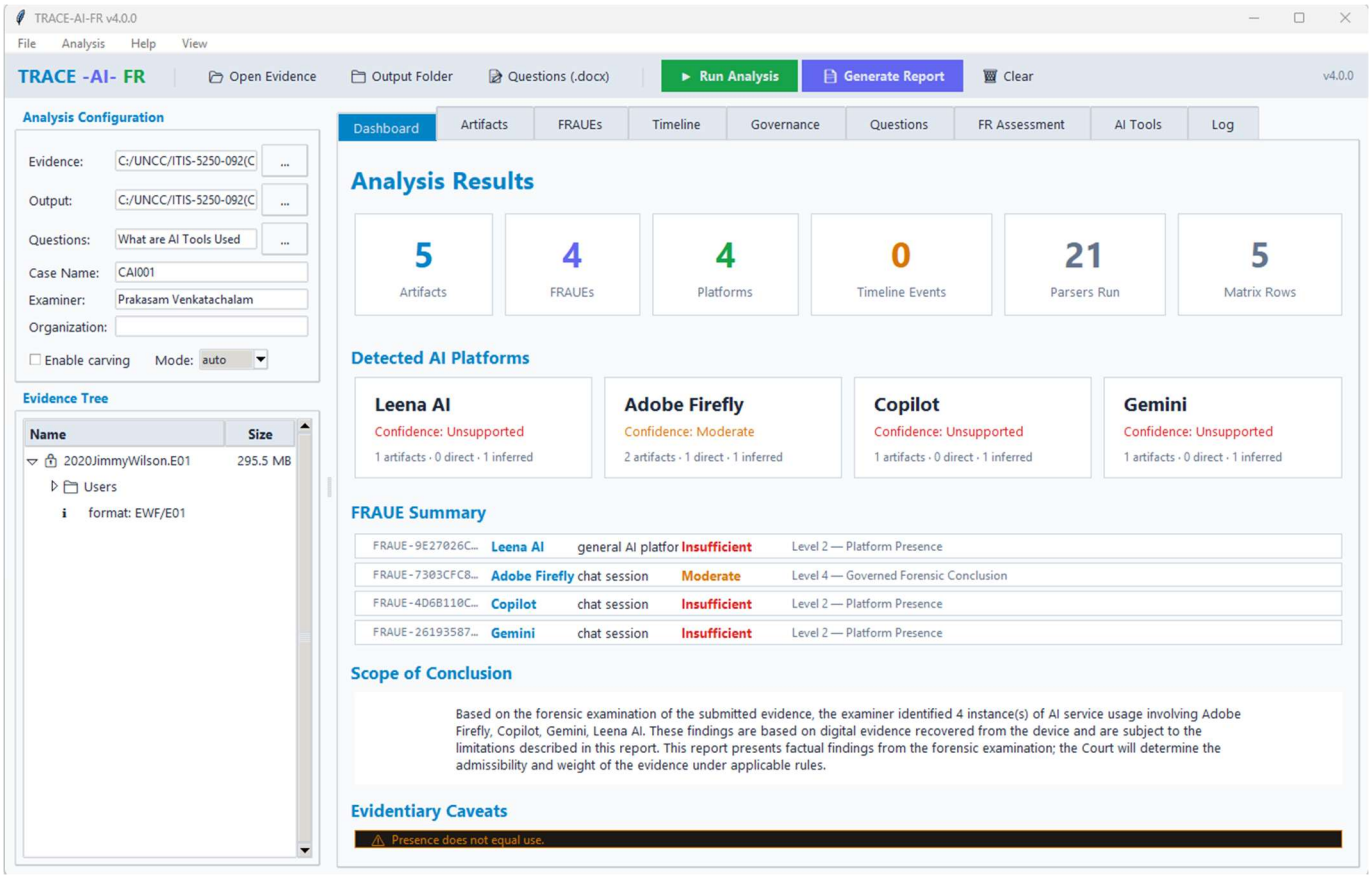
12. Simple First-Test Values

Field	Value
Evidence	Extracted 2020JimmyWilson.E01 file path
Output	C:\TRACE-Output
Questions	What are the AI tools used or Leave blank

Field	Value
Case Name	LOCAL-TEST-001
Examiner	Your name
Organization	Your organization
Enable carving	Unchecked
Mode	e01

Refer to the Above Fig. 1 for reference.

Output:



13. Academic Review Notes

TRACE-AI-FR should be reviewed as a forensic reasoning-and-reporting framework. It supports artifact-based interpretation and transparent reporting.

Important limits:

- AI-platform indicators are not automatic proof of user intent.
- Browser traces, cookies, files, and metadata must be interpreted based on source reliability.
- Carved artifacts may require lower-confidence reporting.
- Reports should be reviewed by the examiner before final use.

- The tool supports forensic reasoning; it does not replace examiner judgment.

14. Final Reviewer Checklist

- ☐ I opened the packaged Windows app.
- ☐ I selected an evidence source.
- ☐ I selected the correct mode.
- ☐ I ran the analysis.
- ☐ I opened the output folder.
- ☐ I reviewed at least one generated report.
- ☐ I reviewed the caveats and interpretation limits.